

Interaksi Manusia dan Komputer (IMK)

THE USER INTERFACE (UI) DESIGN PROCESS

Step 10
Internasionalization
and Accessibility

Step 11-12
Create Meaningful Graphics,
Icons, and Images &
Choose the Proper Colors



Tujuan

- **Step 10**
 - Setelah mengikuti materi ini mahasiswa dapat:
 - Mahasiswa/i dapat mampu menerapkan internasionalisasi dan aksesibilitas dalam desain UI.
- **Step 11,12**
 - Setelah mengikuti materi ini mahasiswa dapat:
 - Mahasiswa/i mampu menerapkan Graphics, Icons, Images & Warna yang sesuai dengan kebutuhan

Outline (Step 10)

- Pertimbangan Internasionalisasi
- Lokalisasi
- Aksesibilitas

Outline (Step 11,12)

- Icons & Graphics
- Penggunaan Multimedia
- Pemmasalahan Warna
- Pemilihan Warna

Pertimbangan Internasionalisasi

Internasionalization

- Today the Internet and the market for software are global. They cross endless cultural and language boundaries, each with its own **requirements, conventions, customs, and definitions of acceptability.**
- Comprehension and recall can be enhanced when presented information is **culturally appropriate** (Spyridakis and Fukuoka, 2002).
- **To make a product acceptable worldwide**, it must be internationalized. A system must also be **designed to be usable by an almost unlimited range of people, being accessible to anyone who desires to use it.** The design concepts used to achieve these goals are called **internationalization and accessibility.**

International considerations

- Localization
- Cultural considerations
- Writing text
- Using images and symbols

Accessibility considerations

- Types of disabilities
- Designing for accessibility

Lokalisasi

Localization

- When to do it:
 - When the market includes few or no English speakers.
 - When translation is required by law or by custom.
 - When the widest possible market is desired.
- When not to do it:
 - When the audience already reads English.
- When the cost of retrofitting or rewriting the software is prohibitive.

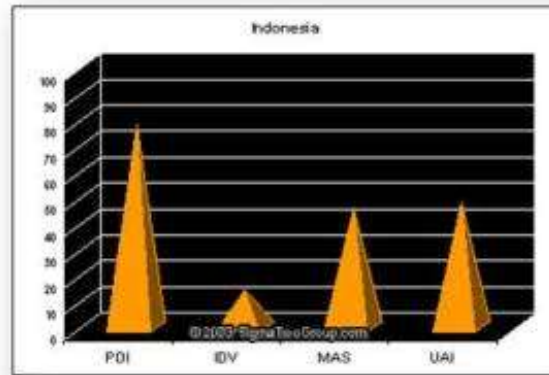


Cultural Considerations

Hofstede (1980, 1997, 2001) has derived and validated five independent dimensions along which cultures vary:

1. **Power distance** is the degree of acceptance of inequality of power and authority
2. **Individualism (vs. collectivism)** reflects how well integrated a person is into the larger group.
3. **Masculinity (vs. femininity)** is the balance between assertiveness or toughness and supportiveness or caring.
4. **Uncertainty avoidance** is the level of discomfort with unstructured and potentially unpredictable conditions.
5. **Long-term orientation (vs. short term)** is the trade-off between long-term reward and immediate recognition.

Indonesia



- PDI(78) → a high level of inequality of power and wealth within the society.
- UAI(48) → more moderated influence of this Dimension within the Indonesian society. In an effort to minimize or reduce this level of uncertainty, strict rules, laws, policies, and regulations are adopted and implemented.
- IDV (14) → the Indonesian society is Collectivist as compared to Individualist.

Words and Text

- Use very simple English
- Avoid
 - Acronyms and abbreviations
 - Slang or obscure phrasing
 - Local or computer jargon
 - A telegraphic/ over friendly writing style
 - Culturally specific examples.
- Adhere to local user language idioms and cultural contexts
- Keep the original term for words that cannot be translated
- Allow additional screen space for the translation
- Position icon captions outside of the graphic
- Modify mnemonics for keyboard access
- Adhere to local formats for date, time, money, measurements, addresses, and telephone numbers



Images and Symbols

- Adhere to local cultural and social norms
- Use internationally accepted symbols
- Develop generic images
- Be particularly careful with
 - Religious symbols (crosses and stars)
 - The human body
 - Women
 - Hand gestures
 - Flags
 - Controversial geographic maps
 - The cross and check for check boxes
- Review proposed graphical images early in the design cycle

Color, Sequence, and Functionality

- Adhere to local color connotations and conventions
- Provide the proper information sequence
- Provide the proper functionality
- Remove all references to features not supported



Cultural Color Associations

	RED	YELLOW	GREEN	BLUE	WHITE
China	Happiness	Birth Wealth Power	Ming Dynasty Heavens Clouds	Heavens Clouds	Death Purity
Egypt	Death	Happiness Prosperity	Fertility Strength	Virtue Faith	Joy
France	Aristocracy	Temporary	Criminality	Truth Freedom Peace	Neutrality
India	Life Creativity	Success	Prosperity Fertility		Death Purity
Japan	Anger Danger	Grace Nobility	Future Youth Energy	Villainy	Death
United States	Danger Stop	Cowardice Caution	Safety Go	Masculinity	Purity

Requirements Determination and Testing

- Establish international requirements at the beginning of product development
- Establish a relationship within the target culture
- Test the product as if it were new



Aksesibilitas

Accessibility

- Accessibility means a system must be usable by an almost unlimited range of people
- Design objectives in creating accessibility for users with disabilities are:
 - Minimize all barriers that make a system difficult, or impossible, to use.
 - Provide compatibility with installed accessibility utilities.
- Types of disabilities:
 - Visual → reduced visual acuity to total blindness.
 - Hearing → inability to detect certain sounds to total deafness.
 - Physical movement → difficulties in, or an inability to, perform certain physical tasks such as moving a mouse, or accurately striking a single keyboard key.
 - Speech or language → difficult to read and write
 - Cognitive → memory impairments and perceptual problems.
 - Seizure disorders → sensitive to visual flash rates, certain rates triggering seizures.

Accessibility Design

- Consider accessibility issues during system planning, design, and testing.
- Provide compatibility with installed accessibility utilities.
- Provide a customizable interface.
- Follow standard Windows conventions.
- Use standard Windows controls.
- Assure online forms can be easily completed.



Visual Disabilities

- Ensure compatibility with screen-review utilities and screen-enlargement utilities
- Screen components:
 - Include meaningful screen and window titles.
 - Provide associated captions or labels for all controls, objects, icons, and graphics.
- Include graphical menu choices.
 - Provide a textual summary for each statistical graphic.
 - Allow for screen element scalability.
 - Support system settings for high contrast for all user interface controls and client area content.
- When a high contrast setting is established, hide any images drawn behind the text to maintain screen information legibility.
- Avoid displaying or hiding information based on the movement of the pointer. Exception: Unless it's part of the standard interface (example: ToolTips).

Visual Disabilities ..

- Keyboard:
 - Provide a complete keyboard interface.
 - Provide a logical order of screen navigation.
- Color:
 - Use color as an enhancing design characteristic.
 - If used, considers
 - color combinations
 - lightness contrast between foreground and background color
 - lightness contrast between colors in the color spectrum (blues and reds)
 - Avoid combining dark colors from the middle of the spectrum with light colors from either end of the spectrum
 - Do not define specific colors
 - Use tools to verify what colors will look like when seen by color-deficient people

Hearing Disabilities



- Provide captions or transcripts of important audio content.
- Provide an option to display a visual cue for all audio alerts.
- Provide an option to adjust the volume.
- Use audio as an enhancing design characteristic.
- Provide a spell-check or grammar-check utility.

Physical Movement Disabilities

- Provide voice-input systems
- Provide a complete and simple keyboard interface
- Provide a simple mouse interface
- Provide on-screen keyboards
- Provide keyboard filters



Cognitive Disabilities



- Permit modification and simplification of the interface.
- Limit the use of time-based interfaces.
 - Do not briefly display critical feedback or messages and then automatically remove them.
 - Provide an option to permit the user to adjust the length of the time-out.

Use elements that do not blink or flicker at rates between frequency ranges of 2 Hz and 55 Hz.

Minimize the area of the screen that is flashing.

Avoid flashing that has a high level of contrast between states.

Provide an option to enable users to slow down or disable screen flashing.

Web Page Accessibility Design

- **WWW Consortium Accessibility Guidelines:**
 1. Provide equivalent alternatives to auditory or visual content
 2. Don't rely on color alone. Image should be understandable
 3. Use markup and style sheets and do so properly
 4. Clarify natural language usage
 5. Create tables that transform gracefully
 6. Pages featuring new technologies transform gracefully
 7. Ensure user control of time-sensitive content changes
 8. Ensure direct accessibility of principles of embedded interfaces
 9. Design for device-independence
 10. Use interim solutions
 11. Use W3C technologies and follow accessibility guidelines
 12. Provide context and orientation information
 13. Provide clear navigation mechanisms
 14. Ensure that documents are clear and simple

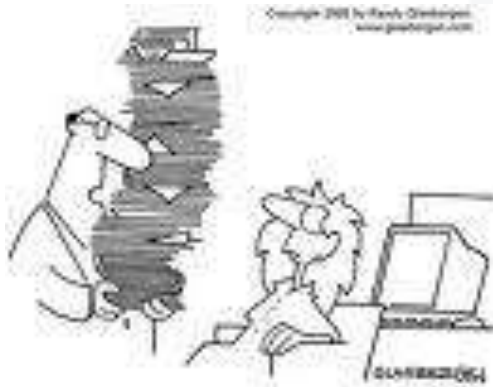
Usability for Lower-Literacy People

- 
- ★ **Prioritize information**
 - ★ **Avoid moving or changing text**
 - ★ **Streamline the page design**
 - ★ **Simplify navigation**
 - ★ **Optimize search**

Usability for Senior Citizens

- Provide large targets to make selection easier
- To reduce the number of selections or clicks,
 - Do not require double-clicks or pull-down menus
 - Do not have a deep page hierarchy.
- Concentrate important information at the top of pages.
- Avoid the need to scroll to find information.
- Put most links in a bulleted, not tightly clustered, list.
- Provide differentiation between visited and non-visited links.
- Use few colors, and avoid using blue and green tones.
- For text to be read or scanned, use
 - 12 to 14 point sans serif font (Helvetica, Arial).
 - Black text on a white background, left justification.
 - Increased spacing (leading) between lines.
 - Sentence style mixed-case letters rather than all capital letters.
 - Appropriate large headings in a 14 to 16 point sans serif font.

Documentation



- Provide documentation on all accessible features
- Provide documentation in alternate formats
- Provide online documentation for people who have difficulty reading or handling printed material

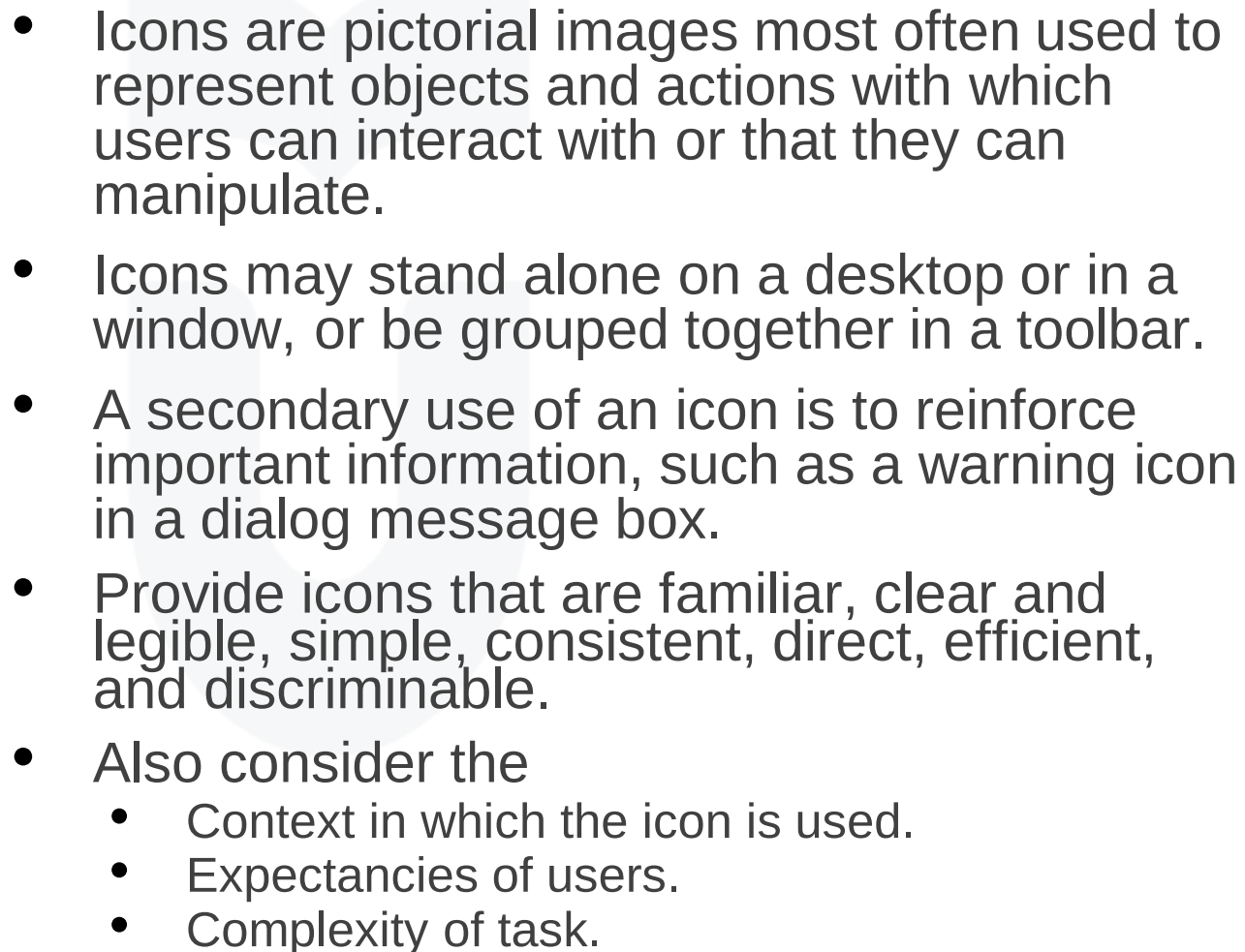


Testing

Test all aspects of accessibility as part of the normal system testing process

Icons & Graphics

#InformatikaBerkarakter



Kinds of Icons

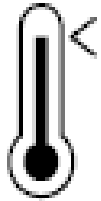
Marcus (1984):

- **Icon.** Something that looks like what it means.
- **Index.** A sign that was caused by the thing to which it refers.
- **Symbol.** A sign that may be completely arbitrary in appearance.

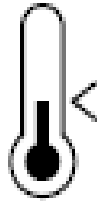
Rogers (1989):

- **Resemblance.** An image that looks like what it means.
Ex. A book → a dictionary
- **Symbolic.** An abstract image representing something.
Ex. A cracked glass → something fragile
- **Exemplar.** An image illustrating an example or characteristic of something. Ex. A freeway exit picturing knife and fork → restaurant
- **Arbitrary.** An image completely arbitrary in appearance whose meaning must be learned. Ex. Menu and sizing icons on screens
- **Analogy.** An image physically or semantically associated with something. Ex. a wheelbarrow full of bricks → move command

Some common icons. What do they stand for?



Hot



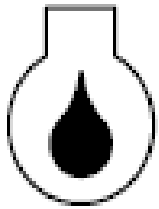
Cold



Fast



Slow



Engine Oil



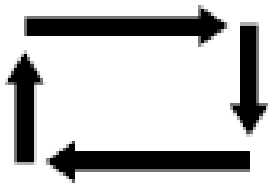
Ammeter/ Generator



Straight



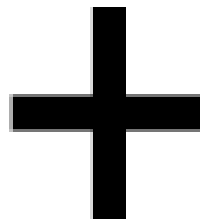
Turn



Automatic



Variable Regulation



Plus/ Positive



Minus/ Neg

Max Number of Codes for Effective Human Differentiation

Encoding Method	Recom. Max.	Comments
Alphanumeric	Unlimited	High versatile. Meaning usually self-evident. Location time may be longer than for graphic coding.
Geometric Shapes	10-20	High mnemonic value. Very effective if shape relates to object or operation being represented
Size	3-5	Fair. Considerable space required. Location time longer than for colors and shapes.
Line Length	3-4	Will clutter the display if many are used.
Line Width	2-3	Good.
Line Style	5-9	Good.
Line Angle	8-11	Good in special cases (Such as wind direction)
Solid and Broken Lines	3-4	Good
Number of Dots or Marks	5	Minimize number for quick assimilation.

Max Number of Codes for Effective Human Differentiation ..

Encoding Method	Recom. Max.	Comments
Brightness	2-3	Creates problems on screens with poor contrast.
Flashing/ blinking	2-3	Confusing for general encoding but the best way to attract attention. Interacts poorly with other codes. Annoying if overused. Limit to small fields.
Underlining	No data	Useful but can reduce text legibility.
Reverse Polarity	No data	Effective for making data stand out. Flicker easily perceived in large areas.
Orientation (location on display surface)	4-8	-
Color	6-8	Attractive and efficient. Short location time. Excessive use confusing. Poor for the color blind.
Combinations of codes	Unlimited	Can reinforce coding but complex combinations can be confusing.

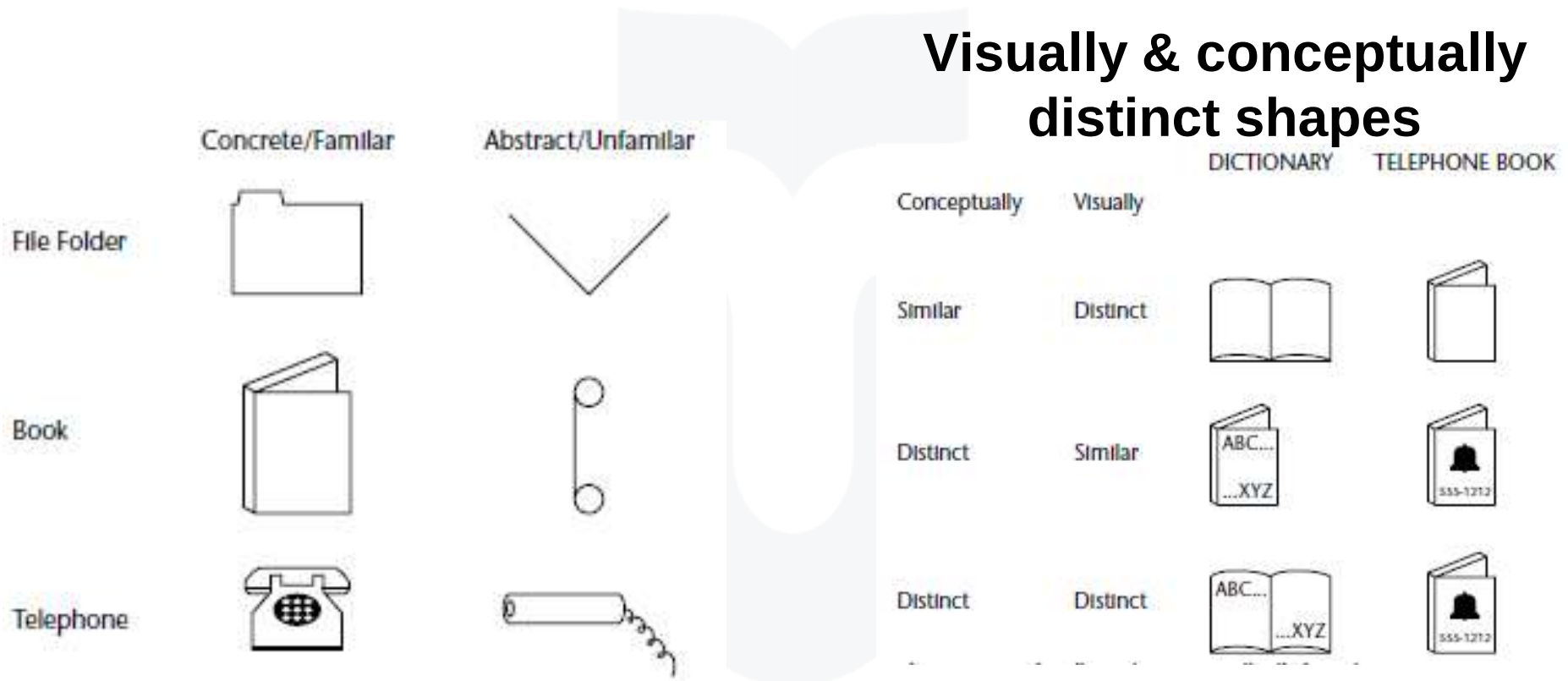
A Successful Icon

- Looks different from all other icons.
- Is obvious what it does or represents.
- Is recognizable when no larger than 16 pixels square.
- Looks as good in black and white as in color.

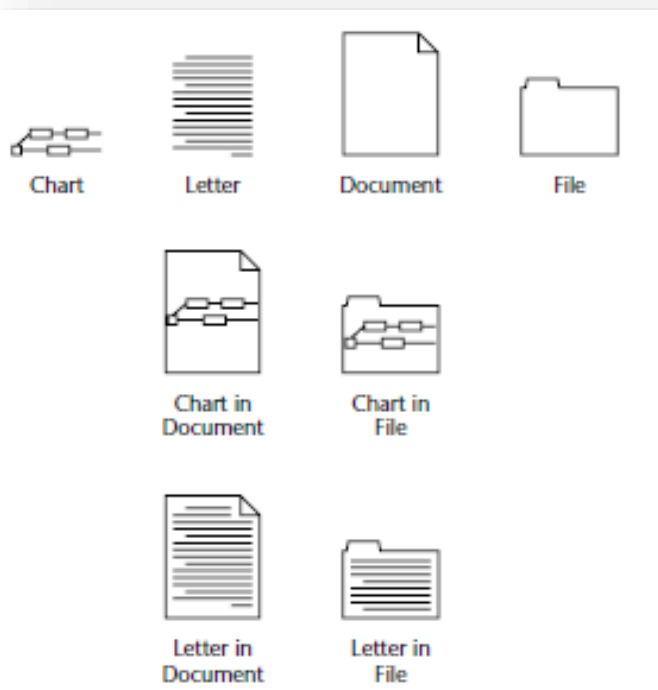
Choosing Icons Images:

- Use existing icons when available.
- Use images for nouns, not verbs.
- Use traditional images.
- Consider user cultural and social norms.

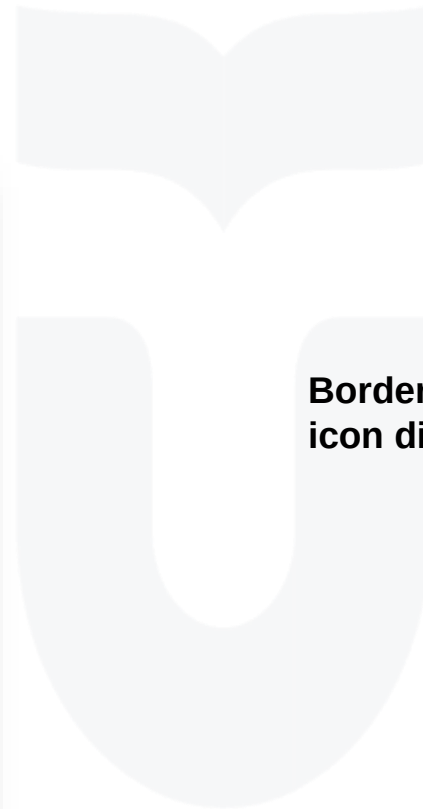
Concrete and familiar shapes



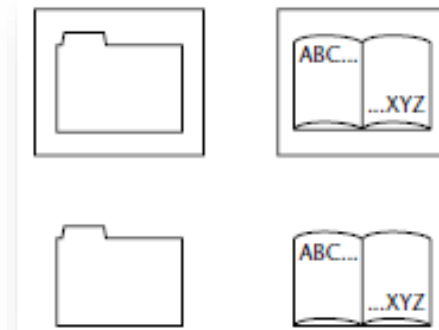
Examples



**Communication relationships
in icons**



**Borders degrading
icon distinctiveness**



**Avoid excessive detail
in icon design**

Penggunaan Multimedia

Multimedia

- The graphical flexibility of the Web permits inclusion of other media on a screen, including images, photographs, video, diagrams, drawings, and spoken audio.
- On the one hand, the various media can be powerful communication and attention-getting techniques.
- On the other hand, effective use of multimedia in design has been hindered by a lack of knowledge concerning how the various media may best be used, and a scarcity of applied design guidelines.
- The objective is **good interaction design**, not “sparkle.”

- Use graphics to
 - Supplement the textual content, not as a substitute for it.
 - Convey information that can't be effectively accomplished using text.
 - Enhance navigation through
 - Presenting a site overview.
 - Identifying site pages.
 - Identifying content areas.
- Limit the use of graphics that take a long time to load.
- Coordinate the graphics with all other page elements.
- Graphics should not look like gratuitous decorations or banner ads.
- Types of graphics (by purpose):
 - **Navigational.** To identify links that may be followed
 - **Representational.** To illustrate items mentioned in text
 - **Organizational.** To relate items mentioned in text
 - **Explanative.** To show how things or processes work
 - **Decorative.** To provide visual appeal and emphasis

Graphics

- **Images:** should convey their intended messages
- **Image maps:** to provide navigation links to other content
- **Photographs/ Pictures:** used when every aspect of the image is relevant
- **Video:** to show things moving/ changing over time
- **Diagrams:** to show the structure/ relationship of objects
- **Drawings:** when selected parts need to be emphasized/ represented
- **Animation:** to explain ideas involving a change in time/ position
- **Audition:** as a supplement to text and graphics

Combining Mediums

- Use sensory combinations that work **best together**
- Closely **integrate** screen text with graphics
- Both the visual and auditory information should be totally **relevant to the task** being performed
- Visual and auditory textual narrative should be presented **simultaneously**, or the visuals should precede the narrative by no more than 7 seconds
- To control **attention**, reveal information systematically
 - Limit elements revealed to one item at a time and use sequential revelations for related elements.
- Consider **downloading times** when choosing a media
- Thoroughly **test all graphics** for
 - Legibility
 - Comprehensibility
 - Acceptance

Learning Improvements for Various Media

MEDIUM	PERCENT MORE LEARNING
Hearing spoken text and viewing graphics	91%
Viewing graphics alone	63%
Viewing text and viewing graphics	56%
Hearing spoken text, viewing text, and viewing graphics	46%
Hearing spoken text and viewing text	32%
Viewing text alone	12%
Hearing spoken text alone	7%

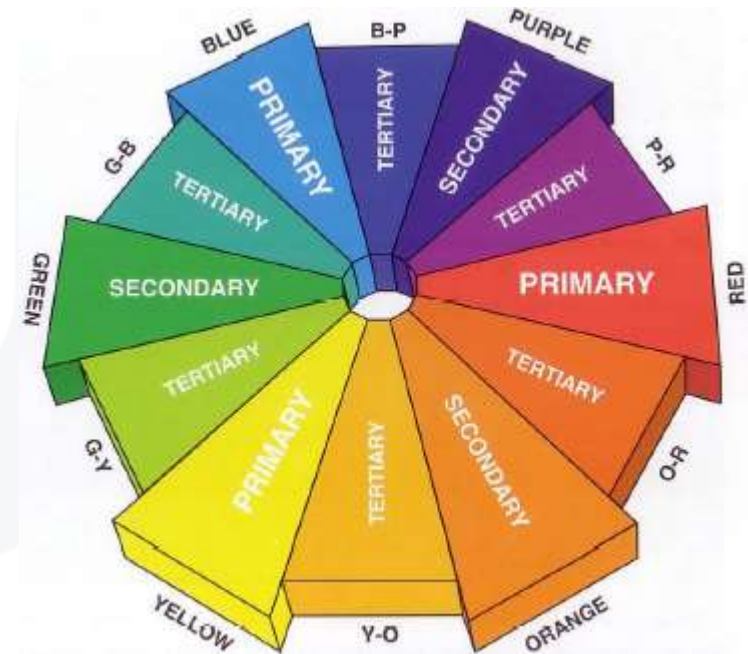
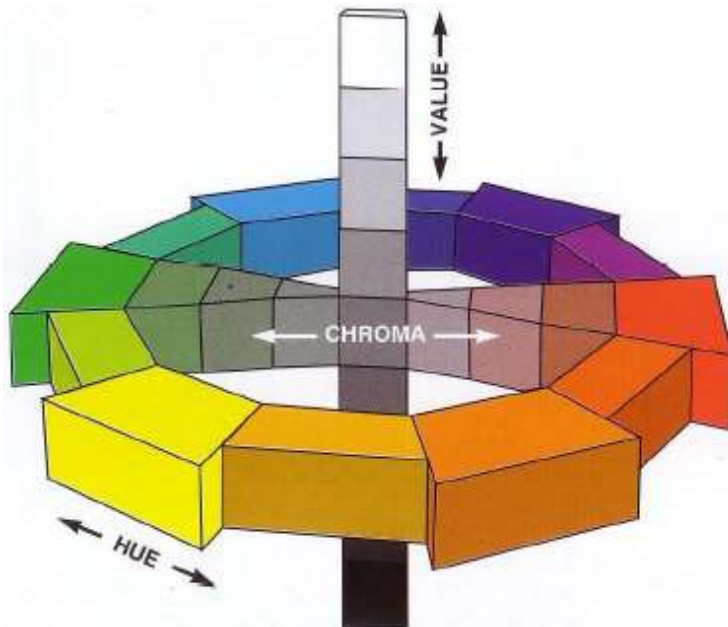
From Lee and Bowers (1997).

Permasalahan Warna

Color

- **Color** adds dimension, or realism, to screen usability.
- Color draws **attention** because it attracts a person's eye.
- If used **properly**, it can emphasize the logical organization of information, facilitate the discrimination of screen components, accentuate differences among elements, and make displays more interesting and attractive.
- If used **improperly**, color can be distracting and possibly visually fatiguing, impairing the system's usability.

Color Dimensions



Color Uses

- Use color to assist in formatting a screen
 - Relating or tying elements into groupings
 - Breaking apart separate groupings of information
 - Associating information that is widely separated on the screen
 - Highlighting or calling attention to important information by setting it off from the other information
- Use color as a visual code to identify
 - Screen components
 - The logical structure of ideas, processes, or sequences
 - Sources of information
 - Status of information
- Use color to
 - Realistically portray natural objects
 - Increase screen appeal

Possible Problems with Color

- High Attention-Getting Capacity
- Interference with Use of Other Screens
- Varying Sensitivity of the Eye to Different Colors
- Color-Viewing Deficiencies

ACTUAL COLOR	COLOR SEEN WITH:		
	RED-VIEWING DEFICIENCY (2.04%)	GREEN-VIEWING DEFICIENCY (6.39%)	BLUE-VIEWING DEFICIENCY (0.003%)
Red	Brown	-	-
Yellow	Greenish-Yellow	Orange	Deeper Yellow
Purple	Dark Blue	Red	Deep Red
Green	-	Light Brown	-
Brown	-	Reddish-Brown	-
Blue	-	-	Green

From Barnett (1993); Fowler and Stanwick (1995).

- Color Connotations
- Cross-Disciplinary and Cross-Cultural Differences

Common Color Connotations

COLOR	POSITIVE	NEGATIVE
Red	Active, Attractive, Dominating, Exciting, Invigorating, Powerful, Strong	Aggressive, Alarming, Energetic
Blue	Abstinent, Controlled, Deep, Dreamy, Faithful, Harmonious, Intellectual, Mysterious, Pornography, Rational, Sensible, Tenderness	Aggressive, Cold, Introverted, Melancholic
Blue-green or turquoise	Refreshing	Aloof, Cold, Self-willed, Sterile, Unemotional
Green	Calm, Close to nature, Conciliatory, Gentle, Harmonious, Optimistic, Refreshing, Strong willed	Envious, Inexperienced, Jealous
Yellow	Cheerful, Colorful, Extroverted, Full of fun, Light, Lively, Youthful	Cowardly, Exaggerated, Superficial, Vain
Orange	Alive, Communicative, Direct, Exciting, Joyful, Warm	Cheap, Intimate, Possessive, Vigorous
Purple	Luxurious, Royal, Serious	Sad

Derived from Gotz (1998) and Stone et al. (2005)

Color – the Research Shows

- Color has been shown
 - to improve performance (Kopala, 1981; Nagy and Sanchez, 1992; Sidorsky, 1982),
 - to improve visual search time (Christ, 1975; Carter, 1982),
 - to be useful for organizing information (Engel, 1980), to aid memory (Marcus, 1986b),
 - to demarcate a portion of a screen (as opposed to lines or type font, Wopking et al., 1985)
- Color:
 - Does not improve performance (Tullis, 1981),
 - Does not affect on reading text (Legge and Rubin, 1986),
 - May impair performance (Christ and T, 1973; Christ, 1975),
 - Is less important than display spacing (Haubner and B, 1983).
- For simple displays, color may have no dramatic impact. As display complexity increases, however, so does the value of color. **To be effective, color must be properly used.**

Effective Foreground / Background Combinations

FOREGROUND	BACKGROUND							
	BLACK	BLUE	GREEN	CYAN	RED	MAGENTA	BROWN	WHITE
BLACK	x			Good		Good		Good
BLUE		x			Poor			Good
H.I. BLUE			Poor	Poor			Poor	Poor
CYAN	Good		Poor	x			Poor	
H.I. CYAN	Good	Good		Good	Good	Good		
GREEN	Good	Good	x	Poor	Good		Poor	Poor
H.I. GREEN		Good						
YELLOW	Good	Good		Good		Good		
RED			Poor		x	Poor	Poor	
H.I. RED			Poor					
MAGENTA			Poor		Poor	x	Poor	
H.I. MAGENTA	Good		Good			Poor		
BROWN			Poor			Poor	x	
GRAY		Poor			Poor		Poor	
WHITE		Good		Poor				x
H.I. WHITE	Good		Good	Good				

(H.I. = High Intensity)
From Lalomia and Happ (1987).

Pemilihan Warna

Choosing Colors for Web Pages

- Color must always have a meaningful purpose
- Use the browser 216-color palette
- Presentation:
 - Minimize the number of presented colors
 - Always consider color in context
 - Use similar or the same color schemes throughout
 - For foregrounds: Use black or strong colors for text and headings
 - For backgrounds: Use weaker contrasting colors such as off-white or light gray
 - Use a uniform color in large areas
 - The smaller the element, the more contrast is required between it and its background
 - Larger images should use
 - Flat, Web-safe colors
 - Fewer colors than small images
 - Select colors to be easily reproduced in black and white
- Use default colors for links
- Do not display non-link text in link colors
- Test all colors

Uses of Color to Avoid

- Relying exclusively on color
- Too many colors at one time
- Highly saturated, spectrally extreme colors together:
 - Red and blue, yellow and purple
- Low-brightness colors for extended viewing or older viewers
- Colors in small areas, or for fine details.
- Colors lacking contrast:
 - Ex. yellow and white; black and brown; reds, blues, brown
- Fully saturated colors for text or other frequently-read screen components
- Pure blue for text, thin lines, and small shapes.
- Non-opponent colors.
- Red and green in the periphery of large-scale displays.
- Adjacent colors that only differ in the amount of blue.
- Single-color distinctions for color-deficient users.
- Using colors in unexpected ways.
- Using color to improve legibility of densely packed text.